

## **Project 2 Report**

### **Exploratory Data Analysis (EDA) on E-Commerce Sales Dataset**

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# 1. Introduction

Exploratory Data Analysis (EDA) is a crucial step in data analytics that helps uncover patterns, trends, anomalies, and relationships within datasets.

The objective of this project is to analyze an e-commerce sales dataset to gain meaningful business insights through statistical analysis and data visualization.

This project was completed as part of **Week 2 Industrial Training at DecodeLabs**, with a focus on transforming raw transactional data into actionable business intelligence.

## 2. Dataset Overview

The dataset contains detailed transactional records of an e-commerce platform.

### Dataset Size:

- **Rows:** 1200
- **Columns:** 14

### Key Features:

- OrderID
- Date
- CustomerID
- Product
- Quantity
- UnitPrice
- Shipping Address
- Payment Method
- Order Status
- Tracking Number
- Items in Cart
- Coupon Code
- Referral Source
- Total Price

The dataset captures multiple aspects of customer purchasing behavior and operational performance.

### 3. Methodology

The analysis was conducted using Python in Jupyter Notebook.

#### Tools and Libraries Used

- Python
- Pandas
- NumPy
- Matplotlib
- Seaborn

#### Steps Performed

##### Data Loading

The dataset was imported into Jupyter Notebook for analysis.

##### Data Inspection

The structure, shape, and summary statistics of the dataset were examined.

##### Data Cleaning

Basic preprocessing was performed:

- Duplicate record checking
- Missing value inspection
- Date format conversion

##### Exploratory Data Analysis

Statistical and visual analysis techniques were applied to identify trends and relationships.

### 4. Analysis

#### 4.1 Statistical Summary

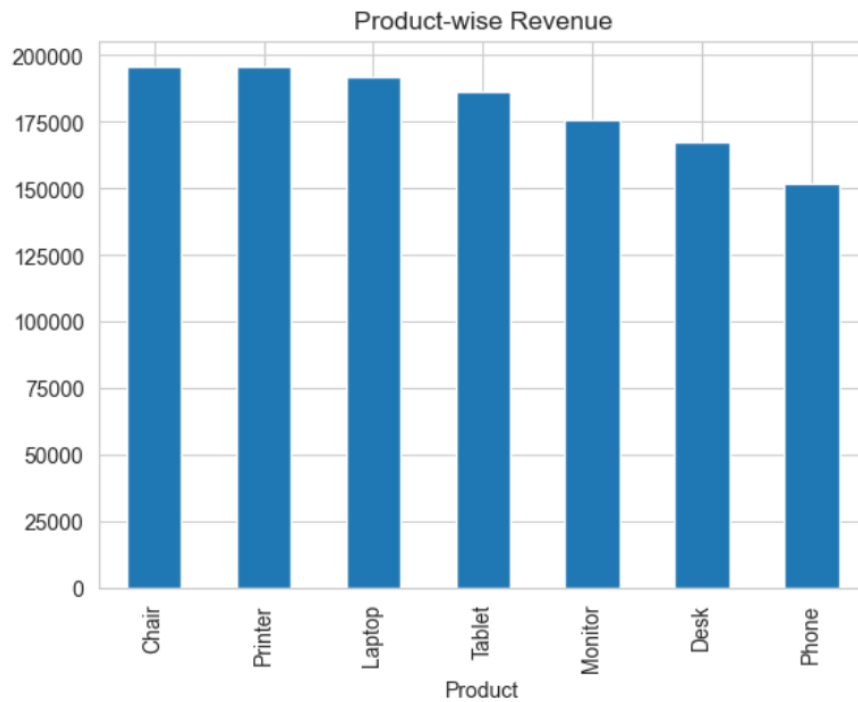
The dataset analysis revealed:

- Average order quantity: approximately **3 units**
- Average unit price: **356**
- Average total order value: **1054**
- Maximum order value: **3456**

This indicates moderate order variability with some high-value purchases.

## 4.2 Product-wise Revenue Analysis

Product-level revenue analysis was conducted to identify high-performing products.



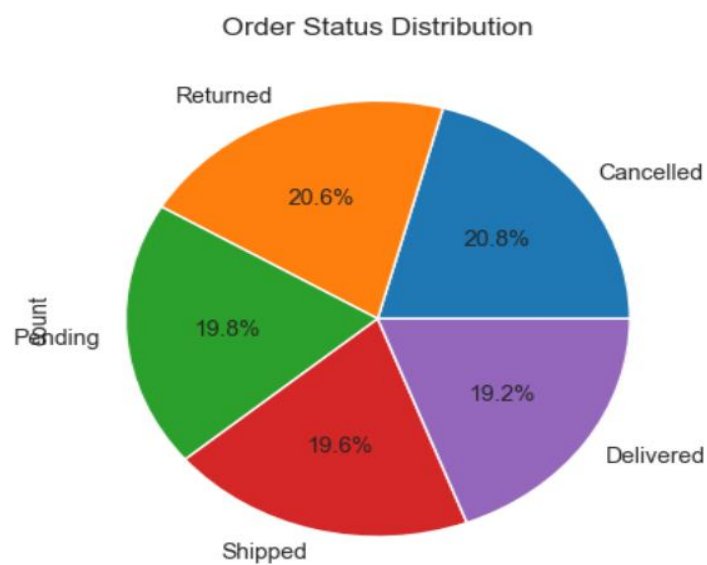
### Observation:

Certain products contribute significantly more revenue than others, indicating uneven product performance.

### Business Insight:

Top-performing products should receive greater promotional focus.

## 4.3 Order Status Distribution



Order statuses were analyzed across:

- Delivered
- Returned
- Cancelled
- Pending
- Shipped

**Observation:**

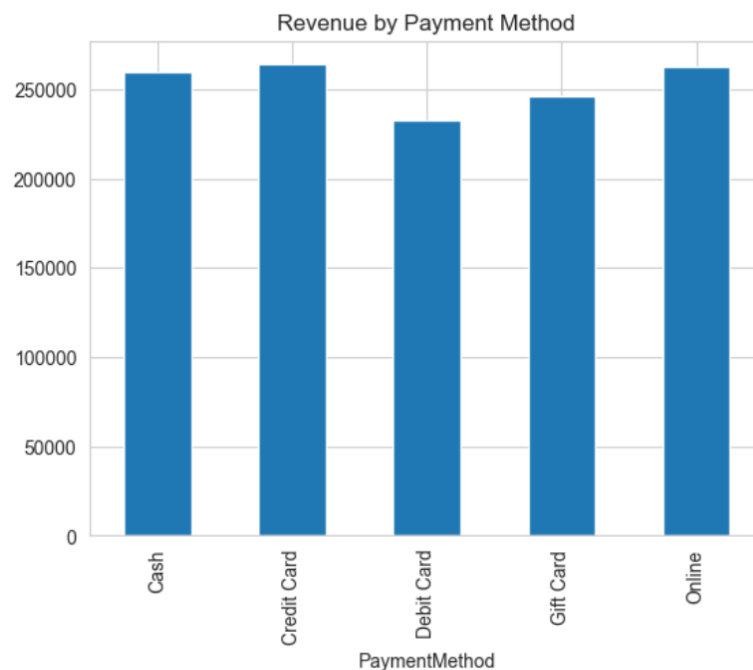
The majority of orders were successfully processed, while a smaller percentage were cancelled or returned.

**Business Insight:**

Returned and cancelled orders should be monitored to identify operational inefficiencies.

#### 4.4 Payment Method Analysis

Revenue contribution by payment method was examined.



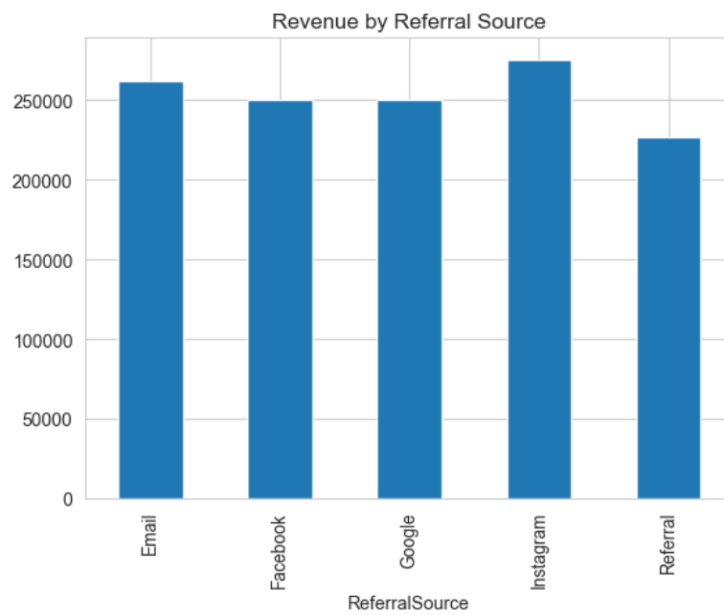
**Observation:**

Digital payment methods contribute strongly to total sales.

**Business Insight:**

Optimizing preferred payment channels can improve customer experience.

## 4.5 Referral Source Analysis



Traffic sources evaluated:

- Instagram
- Facebook
- Email
- Google
- Referral

### Observation:

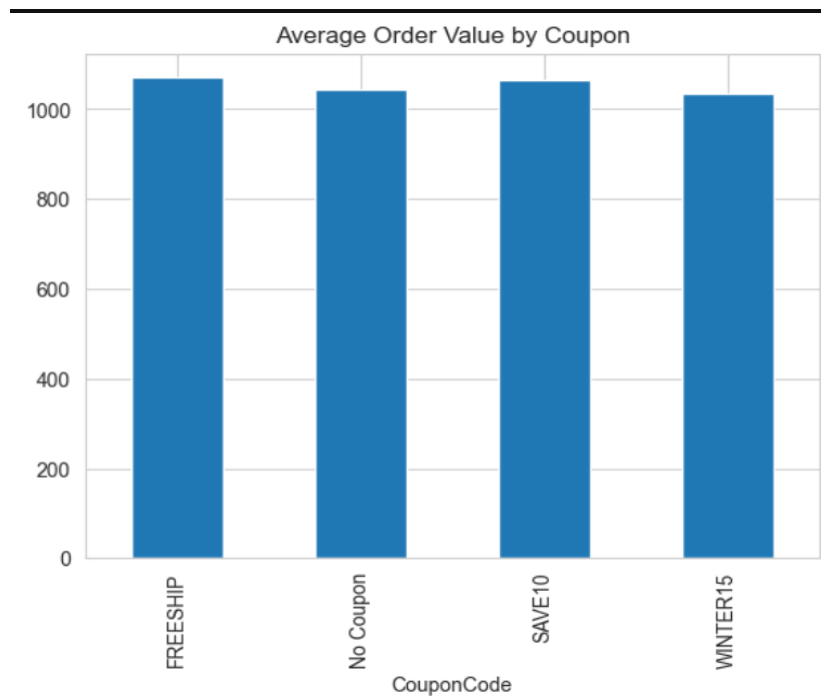
Certain referral channels generate significantly higher revenue.

### Business Insight:

High-performing acquisition channels should receive increased marketing investment.

## 4.6 Coupon Code Analysis

Coupon performance was analyzed using average order value.



**Observation:**

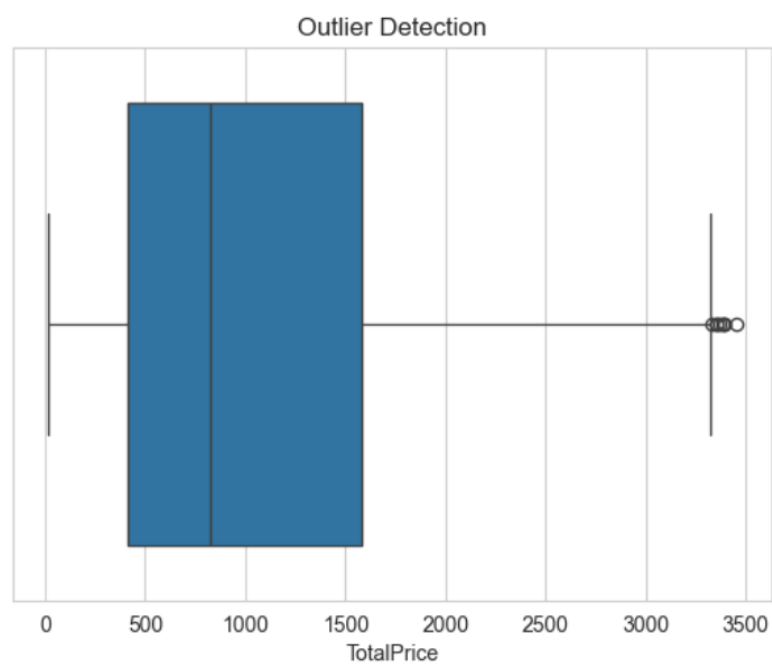
Some coupon codes resulted in higher average purchase amounts.

**Business Insight:**

Effective discount strategies can increase transaction value.

## 4.7 Outlier Detection

Boxplot analysis identified extreme transaction values.



**Observation:**

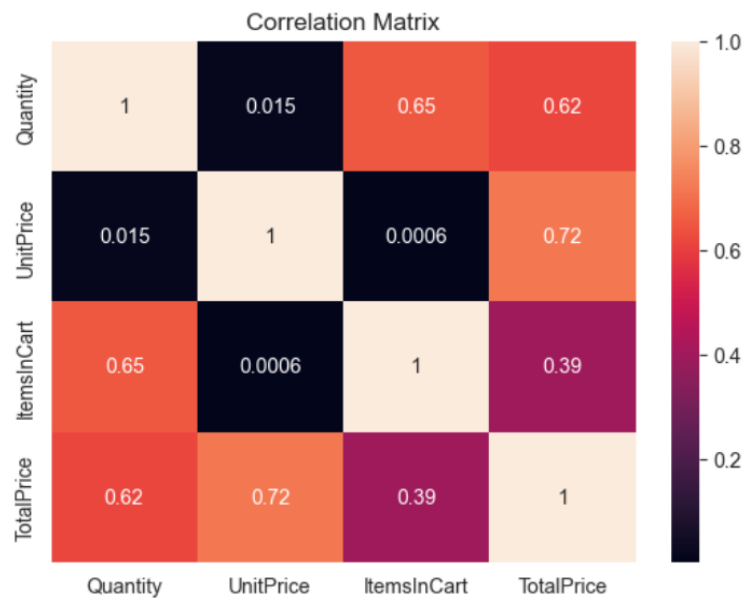
Several unusually high-value transactions were present.

**Business Insight:**

These transactions may indicate premium customer segments or exceptional purchase events.

**4.8 Correlation Analysis**

A correlation heatmap was generated to identify relationships among numerical variables.

**Observation:**

A strong positive relationship exists between:

- Quantity
- Unit Price
- Total Price

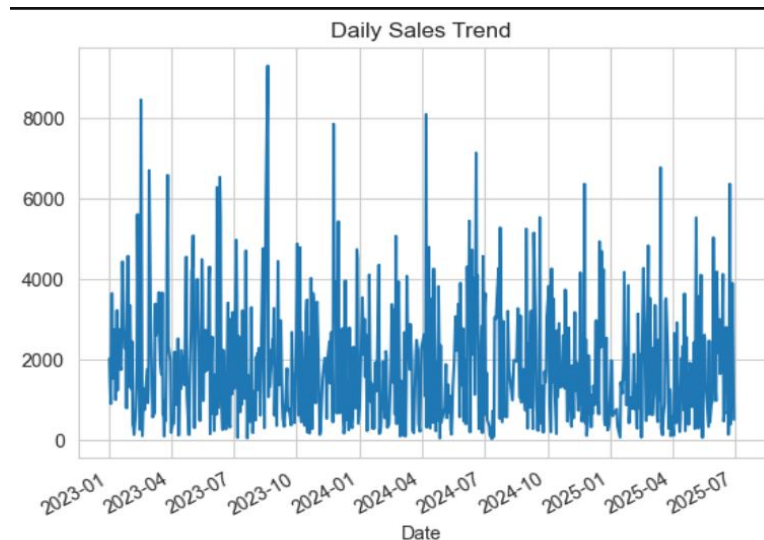
**Business Insight:**

Order quantity significantly impacts revenue generation.

**4.9 Sales Trend Analysis**

Daily sales trends were analyzed using time-series visualization.





**Observation:**

Sales fluctuated across time periods, with distinct peaks and drops.

**Business Insight:**

Seasonal or event-driven sales patterns may influence performance.

## 5. Key Findings

The major findings of this analysis are:

1. The dataset demonstrates consistent transactional activity across products.
2. Revenue generation is concentrated among selected products.
3. Referral channels strongly influence sales performance.
4. Coupon campaigns impact average transaction value.
5. High-value outliers significantly affect revenue distribution.
6. Sales performance varies across time periods.

## 6. Recommendations

Based on the analysis, the following recommendations are proposed:

**Product Strategy**

Focus marketing efforts on top-performing products.

**Customer Acquisition**

Invest more in high-performing referral sources.

**Discount Optimization**

Expand the use of effective coupon campaigns.

### **Operational Monitoring**

Investigate returned and cancelled orders.

### **Revenue Growth**

Leverage high-value customer behavior for targeted strategies.

## **7. Conclusion**

This exploratory data analysis successfully transformed raw e-commerce transaction data into meaningful business insights.

The project demonstrated the practical application of:

- Descriptive statistics
- Data visualization
- Trend analysis
- Correlation analysis
- Business interpretation

This analysis highlights how EDA serves as the foundation for data-driven decision-making.

Through this project, analytical patterns were identified that can support improved business strategy and operational optimization.

This project strengthened practical skills in real-world data analysis and reinforced the importance of converting raw data into actionable intelligence.